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Challenges in landslides risk management in Brazilian slums: a method proposition to prioritizing areas to be mapped in the Municipal Risk Reduction Plan of Niterói, RJ

Os desafios da gestão de risco a movimentos de massa em periferias brasileiras: proposição de método para hierarquização de áreas a serem analisadas no Plano Municipal de Redução de Risco de Niterói, RJ

Franciele Zanandrea¹ , Gean Paulo Michel² , Aimée Guida¹ , Ivanovich Lache Salcedo¹ ,
Danúbia Teixeira Silva³ , Camila Athanázio de Azevedo¹ , Lucas da Silva Santos¹ ,
Isadora Tschiedel Thys¹  & Vanessa Caldas Cardoso¹ 

¹Universidade Federal Fluminense, Niterói, RJ, Brasil

²Universidade Federal do Rio Grande do Sul, Porto Alegre, RS, Brasil

³Instituto Federal de Alagoas, Palmeira dos Índios, AL, Brasil

E-mails: francielez@id.uff.br (FZ), gean.michel@ufrgs.br (GPM), aimeeguida@gmail.com (AG), ilache@id.uff.br (ILS), danubia.teixeira@ifal.edu.br (DTS), azevedocamila@id.uff.br (CAA), lusilva@id.uff.br (LSS), isadorathys@id.uff.br (ITT), vanessa_caldas@id.uff.br (VCC)

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ABSTRACT

In the context of climate emergency, extreme events are expected to become more intense and frequent, disproportionately affecting vulnerable communities. Extreme rainfall events can trigger large-scale and numerous landslides. A landslide risk management approach that focuses not only on response but also on prevention and adaptation becomes crucial. In the city of Niterói - RJ, there are over 100 slums, most of which are located in steep hillslope areas. The method proposed here for creating a qualitative risk index (IR) has been applied and is being validated in the development of the Municipal Risk Reduction Plan of Niterói. The IR combines information on past landslide occurrences and interdictions, vulnerability data, and the susceptibility of the areas —using the SHALSTAB model, to create a ranking of high-risk areas within the slums. Ranking high-risk areas may guide the allocation of technical and financial resources in the field.

Keywords: Landslide; Risk mapping; Disaster risk reduction; SHALSTAB; Vulnerability; Hazard.

RESUMO

Em um cenário de emergência climática, eventos extremos se tornarão cada vez mais intensos e frequentes, afetando de maneira desproporcional comunidades vulneráveis. Eventos chuvosos extremos são capazes de deflagrar deslizamentos de grande magnitude e em grande quantidade, portanto, um gerenciamento de riscos que atue não somente em resposta, mas também em prevenção e adaptação se torna crucial. O município de Niterói abriga mais de 100 comunidades, em sua maioria localizadas em áreas declivosas. O método proposto aqui para a criação de um índice de risco qualitativo (IR) foi aplicado e está sendo validado no desenvolvimento do Plano Municipal de Redução de Risco (PMRR) de Niterói. O IR combina informações sobre histórico de ocorrências e interdições relacionadas a deslizamentos, dados de vulnerabilidade, e de susceptibilidade do terreno - usando o modelo SHALSTAB, para criar uma hierarquização preliminar das áreas de risco dentro das comunidades.

Palavras-chave: Deslizamentos de terra; Mapeamento de riscos; Redução de risco de desastres; SHALSTAB; Vulnerabilidade; Perigo.

INTRODUCTION

Effective landslide risk management faces growing challenges driven by climate-induced extreme events, which are becoming more frequent and more intense amid a global climate emergency (Capobianco et al., 2025; Nogueira & Moura, 2022). The latest IPCC (Intergovernmental Panel on Climate Change) report highlights that human-induced climate change has led to more frequent and intense extreme climatic events, which disproportionately affect economically and socially marginalized urban residents, such as those living in slums (Intergovernmental Panel on Climate Change, 2023). In South America, the key risks identified include floods and landslides as a consequence of the aforementioned extreme events (Intergovernmental Panel on Climate Change, 2023; Seneviratne et al., 2012).

Landslides constitute a hazard with potential to cause extensive socioeconomic and environmental damage, affecting particularly regions with steep topography and vulnerable communities (Capobianco et al., 2025; Coelho-Netto et al., 2009). Extreme rainfall events, capable of triggering large-magnitude and more frequent landslides, have occurred in areas not typically recognized for such processes, such as the mountainous region of the state of Rio Grande do Sul in 2024 (Egas et al., 2025). In the scenarios outlined by the Intergovernmental Panel on Climate Change (2023), with global temperatures rapidly rising, hydrological processes related to the triggering of landslides are also undergoing changes that influence the manner, frequency, and magnitudes of these phenomena. These climate change-driven alterations in the hydrological cycle can vary spatially and temporally in different aspects and magnitudes, consequently affecting landslide triggering in distinct ways, depending on the type of landslide (Jakob, 2022).

In Brazil, landslides are primarily triggered by intense rainfall events, conditioned by other contributing factors such as preexisting soil moisture and changes in slope conditions, usually human-driven (Egas et al., 2025; Ehrlich et al., 2021; Coelho-Netto et al., 2009). According to the Centre for Research on the Epidemiology of Disasters (CRED) international disaster classification, rainfall-triggered landslides are considered hydrological disasters (Below et al., 2009).

The southeastern region of Brazil is one of the most affected by landslides, enduring severe social and economic losses. The steep hillslopes of Rio de Janeiro state are particularly affected, as a consequence of their steep physiography, intense rainstorm pattern and the presence of densely inhabited slums located in susceptible areas (Dias et al., 2021; Instituto Brasileiro de Geografia e Estatística, 2019; Coelho-Netto et al., 2009).

The city of Niterói, located in Rio de Janeiro state, encompasses approximately 117 slums (Niterói, 2024a), most of which in steep hillslope areas. The topographic attributes of these areas make them particularly susceptible to landslides, increasing disaster risks and significantly affecting the quality of life of the residents.

The substantial number of people living in high-risk areas, and the continuous expansion of urbanization into susceptible areas, reveal the need for effective risk management approaches to prioritize investments and planning of structural interventions for risk reduction and mitigation. In 2023, around 80 high-risk areas in Niterói received slope stabilization structural interventions

(Niterói, 2024b). Despite investments in slope stabilization, these efforts remain insufficient to fully mitigate landslide risks in the extension of the susceptible areas with high demographic density due to economic and environmental constraints.

To effectively address landslide risk, governmental agencies need to develop a better understanding of the territory and make technical and objective decisions about the allocation of funds for risk management. Risk mapping and risk assessments are processes underlined by the subjective judgment of the practitioner (Dai et al., 2002), who often times is faced with limitations and uncertainties inherent to the process, such as the variety of risk assessment approaches that can reasonably be adopted, incomplete -or the lack of - historical data recorded by civil defense agencies, restricted access to risk areas in slums with high crime rates, as well as budgetary and personnel constraints. These complicating factors reveal the need for a risk assessment criterion that is not only objective but also spatially distributed, enabling a technical and quantitative assessment of hazards and vulnerabilities across the entire territory of interest.

Municipal Risk Reduction Plans (in Portuguese *Plano municipal de Redução de Risco*, PMRR) are a risk management instrument that aims to establish guidelines for preventive and mitigating disaster actions. Here, the methods that were developed and applied to quantify vulnerability and hazard regarding landslides in urban peripheral occupation areas (slums) of Niterói, as part of the PMRR, are presented. The method is based on the definition of a calculated qualitative Risk Index (IR) to be applied to the territory of interest to create a priority order among the highly susceptible areas. The IR ranking of high-risk areas aim to serve as a technical and objective first approach for risk assessment of the area of interest and can provide decision-makers clear criteria to support the implementation of risk management measures.

METHODS

To promote a better understanding of the territory to be mapped in the future stages of the PMRR, while distinguishing the landslide-prone areas with the highest risk, a method to calculate the Risk Index (IR) was developed. The proposed method consists of an index-based system that assesses various factors influencing the risk but does not explicitly account for probability or potential damage associated with the process.

The IR combines georeferenced data from three main sources to evaluate hazard and vulnerability (Figure 1): the historical record of landslides, residents' financial data and landslide susceptibility mapping of the studied territory.

The landslide historical data encompasses the Civil Defense of Niterói record of landslides and all available registries of civil defense actions recommending house interdictions related to landslides and other geological risks. These historical data went through detailed filtering to exclude all records related to causes other than landslides and associated processes. The vulnerability evaluation was made with a simple approach to identify financially vulnerable areas, regionalizing the parameters through the average family income and the number of residents registered to the CadÚnico. CadÚnico is a nationwide registry for governmental social security programs that records geographical and financial

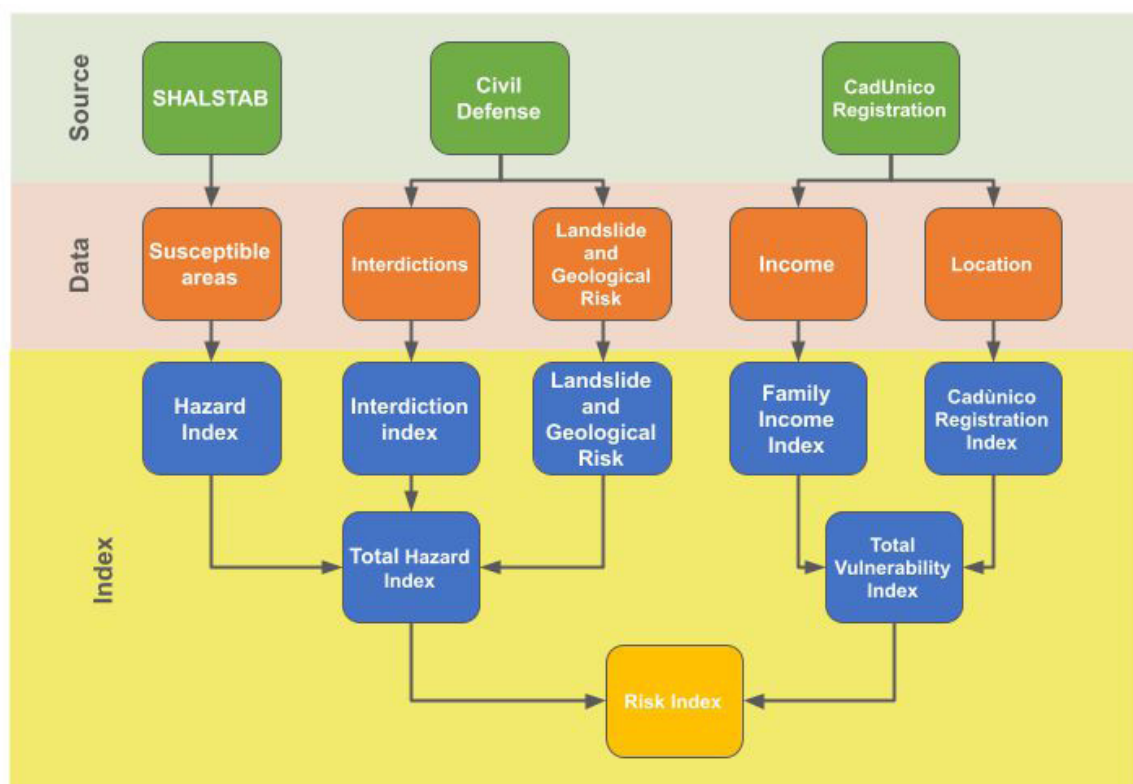


Figure 1. Risk index (IR) calculation process, data, and sources.

data from low-income families. The non-published data used for this analysis were provided for research purposes through a cooperation agreement with the public agency responsible, in accordance with the terms of the General Data Protection Law that guarantees information confidentiality in Brazil (Brasil, 2018). The IR also considers susceptibility, adopting the numerical modeling of physical processes on the slopes and classifying their stability through equations applied in the SHALSTAB (Shallow Landsliding Stability Model) model, proposed by Dietrich & Montgomery (1998).

This multi-source approach made it possible to estimate the risk of the studied areas by calculating the hazard and vulnerability variables. The IR was calculated at two different spatial scales: one covering the mapped slums of Niterói and the other using a regular hexagonal grid extending across the entire municipality. The IR resulted in a priority ranking of the slums to be visited and mapped in the following stages of the PMRR development. Therefore, the IR proposed to evaluate risk in Niterói underwent a validation process that involved a fieldwork phase. The fieldwork phase included participatory mapping, adopting methodology adapted from Virgens et al. (2024), and risk assessment.

Study area

Niterói municipality, in the state of Rio de Janeiro, is located on the eastern margin of Guanabara Bay, encompassing an area of circa 129.4 km² (Figure 2). Niterói has a population of 481,749 inhabitants and a population density of 3,601.67 inhabitants/

km² (Instituto Brasileiro de Geografia e Estatística, 2022). There are approximately 117 slums in Niterói, out of which, the majority are located in steep hillslopes, increasing the exposure of their residents to landslide hazards.

Niterói's geomorphological setting is associated with the lithologies and structures that underlain the city and were shaped through geomorphological processes that originated the mountainous domain, hills, coastal massifs, and plains that compose the city's relief. The geology of the area is primarily composed of Precambrian igneous and metamorphic rocks, as well as Quaternary sediments. The gneisses and granites, which formed in the context of the Ribeira belt and were affected by the Mesozoic tectonic event that originated the Atlantic Ocean, sustain the elevated areas of the city's relief (Suarez, 2005; Dantas & Costa, 2017). The hot and wet climate, through chemical weathering, favors the occurrence of saprolite-mantled hillslopes. On steep mountain slopes and coastal massifs, the lack of stability leads to the exhumation of the bedrock escarpments.

Climate-wise, Niterói is influenced by its proximity to the ocean, where a tropical humid climate prevails. The average annual temperature is 25.2 °C (Roriz, 2013). The rainy season spans from December to March, with an annual average precipitation of approximately 1,200 mm.

Throughout its history, the city of Niterói has faced several disasters related to landslides. The municipality's susceptibility to this kind of occurrence is evident when examining the historical record of major landslide-related disasters, that occurred in 2010 (Morro do Bumba, 48 fatalities), 2013 (Morro do Palácio, 1 fatality), 2018 (Morro da Boa Esperança, 15 fatalities), and the most

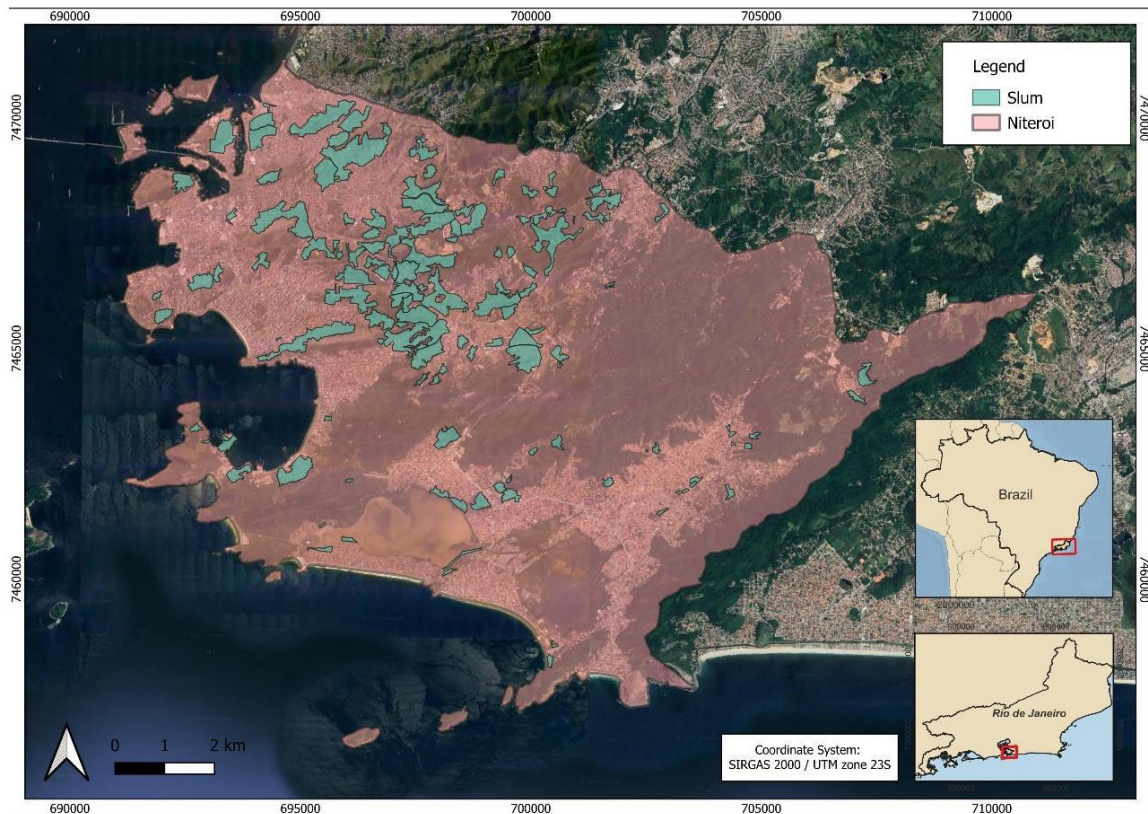


Figure 2. Study area: Map of Niterói municipality, in Rio de Janeiro State, showing the delimitation of the slums of the city.

recent disaster recorded in Jardim Viçoso in January 2024 - in the same neighborhood where Morro do Bumba is located. During this last-mentioned event, intense rainfall triggered a landslide, but fortunately, there were no fatalities.

SHALSTAB

SHALSTAB is a deterministic model based on the combination of an infinite slope stability geomechanical theory and a steady-state hydrological model (Dietrich & Montgomery, 1998). The model analyzes the hydrological conditions necessary for a specific slope to collapse, as a function of a hydrologic ratio (q/T), according to Equation 1.

$$\frac{q}{T} = \frac{b}{a} \sin \theta \left\{ \frac{\rho_s}{\rho_w} \left(1 - \frac{ta\theta}{\tan \phi} \right) + \frac{c}{\cos^2 \theta \tan \phi \rho_w g z} \right\} \quad (1)$$

where: q is the critical uniform rainfall (m/h), T is the soil transmissivity (m^2/h), a is the upstream contributing area of the analyzed point (m^2), b is the size of the pixel being analyzed (m), θ is the slope angle ($^\circ$), ρ_s is the wet soil bulk density (kg/m^3), ρ_w is the density of water (kg/m^3), ϕ is the internal friction angle of the soil ($^\circ$), c is the cohesion (Pa), and z is the soil depth (m). The input parameters required are c , ρ_s , ϕ , and z , and the other variables a , b and θ are extracted from a Digital Elevation Model (DEM). SHALSTAB was selected for this study due to its ability

Table 1. SHALSTAB Input Soil Mechanic Parameters and determined stability threshold.

Soil Mechanic Parameters	Value
Cohesion (c)	2 500 Pa
Soil depth (z)	1 m
Angle of internal friction (ϕ)	31°
Bulk density (ρ_s)	1 700 kg/m^3
$\log q/T$ threshold	-3.1

to quickly produce reliable results using a deterministic approach (Michel et al., 2014).

A DEM with a 0.5m resolution, derived from LiDAR survey was adopted in the modelling. Soil parameters (Table 1) were obtained through model optimization based on previously mapped landslide scars in the municipality (Vasconcelos et al., 2024). The optimization process involved defining parameter ranges and randomly combining values to identify the set that produced the best match between the simulated unstable areas and the observed landslide scars. The range of the parameters, as well as their optimized values, was established and is consistent with the work of Mendonça (2017). In the case of the municipality of Niterói, most of the slopes susceptible to landslides are composed of residual soils derived from orthogneisses, which leads to an expectation of low variability in parameters along the slopes within the study area. In municipalities or regions where high variability

in geotechnical parameters is expected, the optimization process should ideally be carried out for each geotechnical unit individually.

Risk analysis scales

For the risk analysis, two scales (i.e., polygon types) were defined to approach the study area: the polygon of the slums, as delimited by the Niterói governmental agencies (Niterói, 2024a, Figure 3), and a regular hexagonal grid (honeycomb, Figure 3). The hexagonal grid was added to the entire territory, aiming to divide the city into equal areas, enabling a direct comparison between the polygons, as the slums have different territorial extensions. The employment of hexagonal lattices has several advantages to the survey and sampling of a geospatial framework, such as reduced bias from edge effects, better connectivity and pathways analysis, and greater clarity for data visualization (Birch et al., 2007).

The size of the hexagon was defined to approximately match the area of the smallest mapped slum in Niterói. This

hexagonal grid divided the territory into 6,270 hexagons, with sides (L , Figure 3) of approximately 133 meters, resulting in an area of 46,174 m² per hexagon.

After the analysis scale was defined, the risk index (IR) was determined as a product of hazard and vulnerability indices for each studied feature level. All indices are proportional to the parameter they represent, thus, the higher the index, the greater the hazard or the vulnerability. The indices were normalized to allow direct comparisons on the same scale.

Risk index

As previously defined, the IR of a given polygon (hexagon or slum) is calculated as the product of the hazard and vulnerability indices determined for that polygon. The equations used to determine each one of the indices can be found in Table 2.

Three hazard indices were defined, based on the following information: 1) Areas susceptible to landslides as identified through

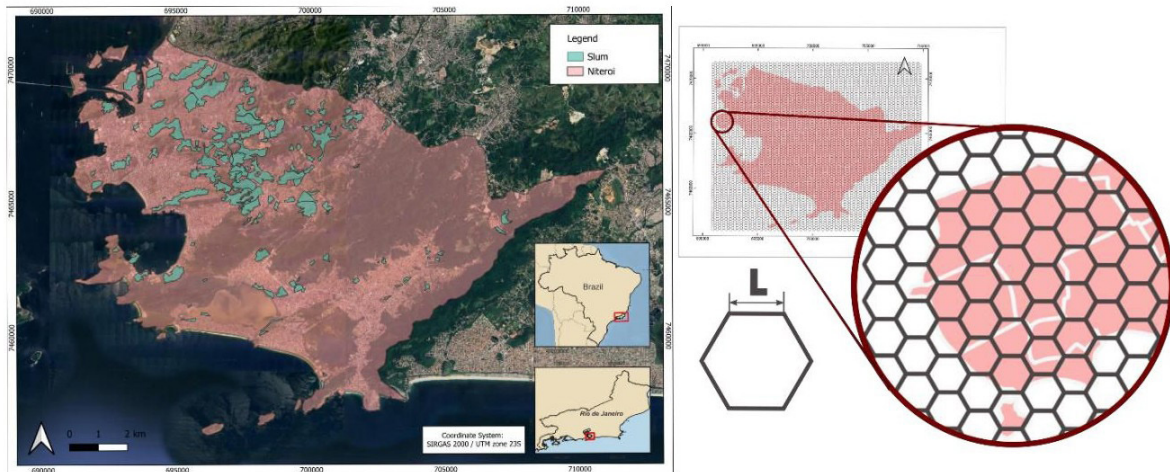


Figure 3. a) Slums of Niterói city. b) Detail of hexagonal grid dividing Niterói territory in equal area polygons (honeycomb).

Table 2. Equations used to calculate Risk, Vulnerability, and Hazard indices.

Hazard Index SHALSTAB (I_{HM})	Landslide and Geological Risk Index (I_{LR})	Interdiction Index (I_I)	Total Hazard Index (I_{TH})
$I_{HM} = \frac{A_H}{A_P}$	$I_{LR} = \frac{N_{LGR}}{A_P}$	$I_I = \frac{N_{IR}}{A_P}$	$I_{TH} = \frac{I_{HM} + I_{LR} + I_I}{3}$
A_H : Area identified as hazardous	N_{LGR} : Total number of landslides and geological risks recorded in the polygon	N_{IR} : Total number of interdictions recorded in the polygon	I_{HM} : Hazard Index (SHALSTAB)
A_P : Polygon area	A_P : Polygon area	A_P : Polygon area	I_{LR} : Landslide and geological risk index
			I_I : Interdiction index
CadÚnico Register Index (I_{CR})	Family Income Index (I_{FI})	Total Vulnerability Index (I_{TV})	Risk Index (I_R)
$I_{CR} = \frac{N_{CR}}{A_P}$	$I_{FI} = \frac{M_{FI}}{A_P}$	$I_{TV} = \frac{(1 - I_{FI}) + I_{CR}}{2}$	$I_R = I_{TH} * I_{TV}$
N_{CR} : Total number of people registered in CadÚnico in the polygon	M_{FI} : Mean family income in the polygon area	I_{FI} : Family income index	I_{TH} : Total hazard index
A_P : Polygon area	A_P : Polygon area	I_{CR} : CadÚnico register index	I_{TV} : Total vulnerability index

the SHALSTAB model (I_{HM}); 2) Historical records of landslides and geological risk from the Civil Defense (I_{LR}); and 3) Historical records of interdictions carried out by the Civil Defense, related to landslides and associated geological processes (I_D). Each Index was calculated according to the equations presented (Table 2). The Total Hazard Index (I_{TH}) derives from these three hazard indices.

To address vulnerability, two indices were defined: 1) the CadÚnico Records Index (I_{CR}), which evaluates the number of people registered to the social security government registry per analyzed polygon, and 2) the Family Income Index (I_{FI}), that encompasses the mean family income of the low-income families in the analyzed polygon.

These indices are used to determine a Total Vulnerability Index (I_{VI}), according to the equation established (Table 2). The risk index (IR) is, then, calculated as the product of the Total Vulnerability Index (I_{VI}) and the Total Hazard Index (I_{TH}).

The risk index calculated values were applied to create a ranking of priority slums to be mapped in the following stages of the PMRR, based on the operational constraints of the team. For effective results, it is necessary that the team applying the method evaluates their own operational limitations, if any.

A total area of 1 km² was defined to limit the surveys and risk assessments in the field. Based on this operational constraint, the IR threshold value was defined for each type of polygon. The IR ranking underwent manual refinement, where slums that had already been granted two or more slope stabilization works were

removed from the IR ranking, creating space for other high-risk polygons to be considered.

RESULTS AND DISCUSSIONS

Landslide susceptibility and vulnerability mapping

The landslide susceptibility of Niterói was estimated through the stability map derived from the SHALSTAB model. The results of the simulation describe the susceptibility to shallow landslides in terms of $\log q/T$, classifying stability in a continuous scale from unconditionally stable (light blue) to unconditionally unstable (reds, Figure 4). The topographic attributes of the terrain, such as slope and contributing area, are incorporated in the calculations of stability for each analyzed cell. These attributes derive from the Digital Elevation Model.

Naturally, the identified unstable areas concentrate along the hillslopes and are influenced by the steepness of the slope and flow lines of their contributing areas. There is a strong correspondence between rock outcrops and areas classified as Unconditionally Unstable, as also observed in Montgomery & Dietrich (1994). The slums in Niterói are predominantly situated on such hillslopes, meaning most of the slums are susceptible, to some degree, to landslide hazards. To rank the slums according to their greater risk, mathematical indices were applied. The stability results from

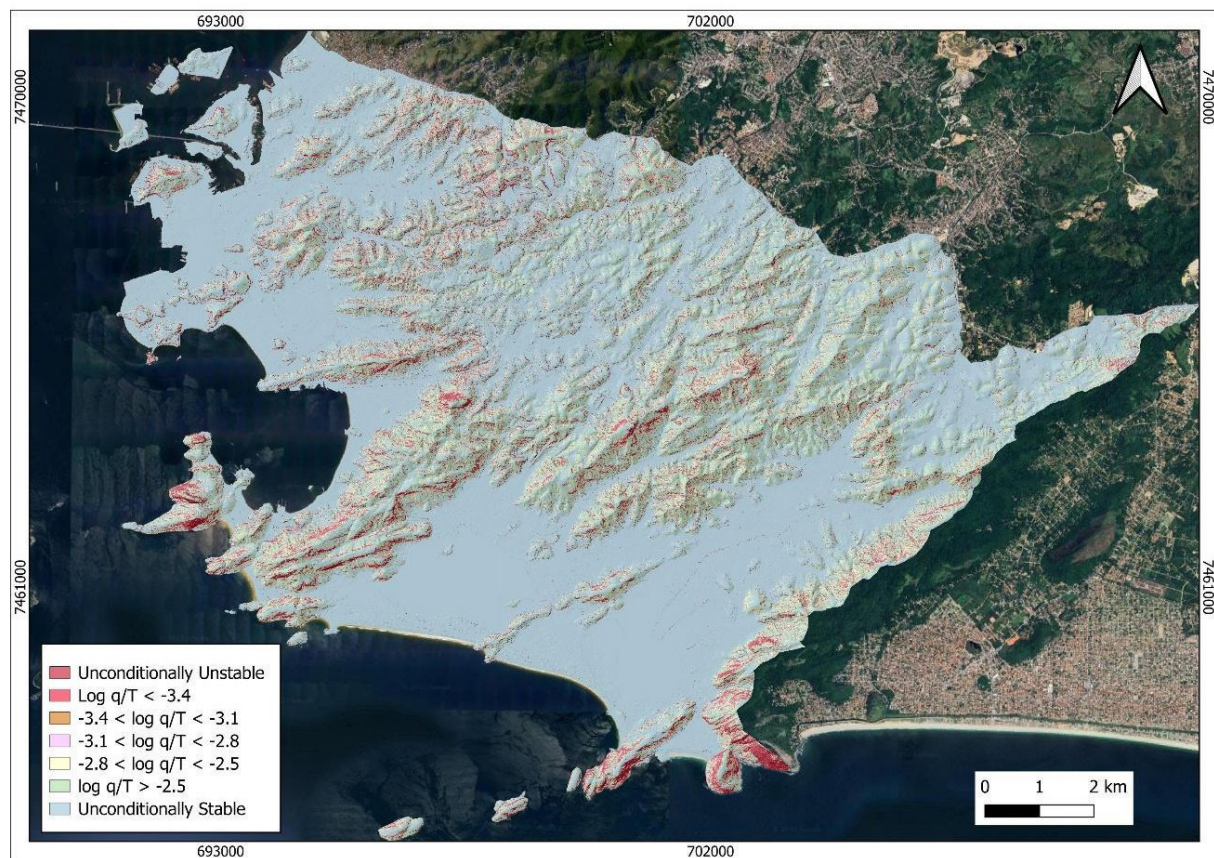


Figure 4. SHALSTAB model stability map result for Niterói territory, classified by stability classes that range from unconditionally stable, in blue, to unconditionally unstable, in red. The percentage of the municipality's area falling into each stability class, from most unstable to most stable, is 7.1%, 0.9%, 0.8%, 1.5%, 2.9%, 22.6%, and 64.3%.

the deterministic model were used to calculate the Hazard Index (I_{HM}), by defining the proportion of areas identified as unstable to the total area of the polygon (hexagon or slum). This index was further improved by incorporating the historic record of landslides and related interdictions from the civil defense database through the Landslides (I_{LR}) and Interdiction (I_I) indices, to create the Total Hazard Index (I_{TH} , Figure 5) for each analyzed scale.

This proposed multi-scale approach of the method allows for a better understanding of the slum's territory, granting the identification of critical areas at risk within slums with larger territorial extents (Figure 5). This is an advantageous characteristic that supports the optimization of decision-making by guiding the allocation of technical and financial resources in the field.

To address vulnerability on a large scale, encompassing the whole territory of Niterói, the same multi-scale approach was

applied, analyzing the slums and the hexagonal grid. Vulnerability is a multifactorial complex concept with no universal method of quantification (e.g., Ávila et al., 2024; Galli & Guzzetti, 2007) and simplifications of the concept were employed in the method to allow the available data to be factored in the calculation of risk. The understanding that landslides affect more severely and cause disproportionate damage to the poorest territories (Intergovernmental Panel on Climate Change, 2023) led to the employment of two vulnerability indices, in both analyzed scales, to consider average family income (I_{FI}) and the number of people registered on the national social security programs database CadUnico (I_{CR}), that resulted in a preliminary classification for the most vulnerable slums (Figure 6). The Hazard and vulnerability indices were then combined to calculate the Risk Index.

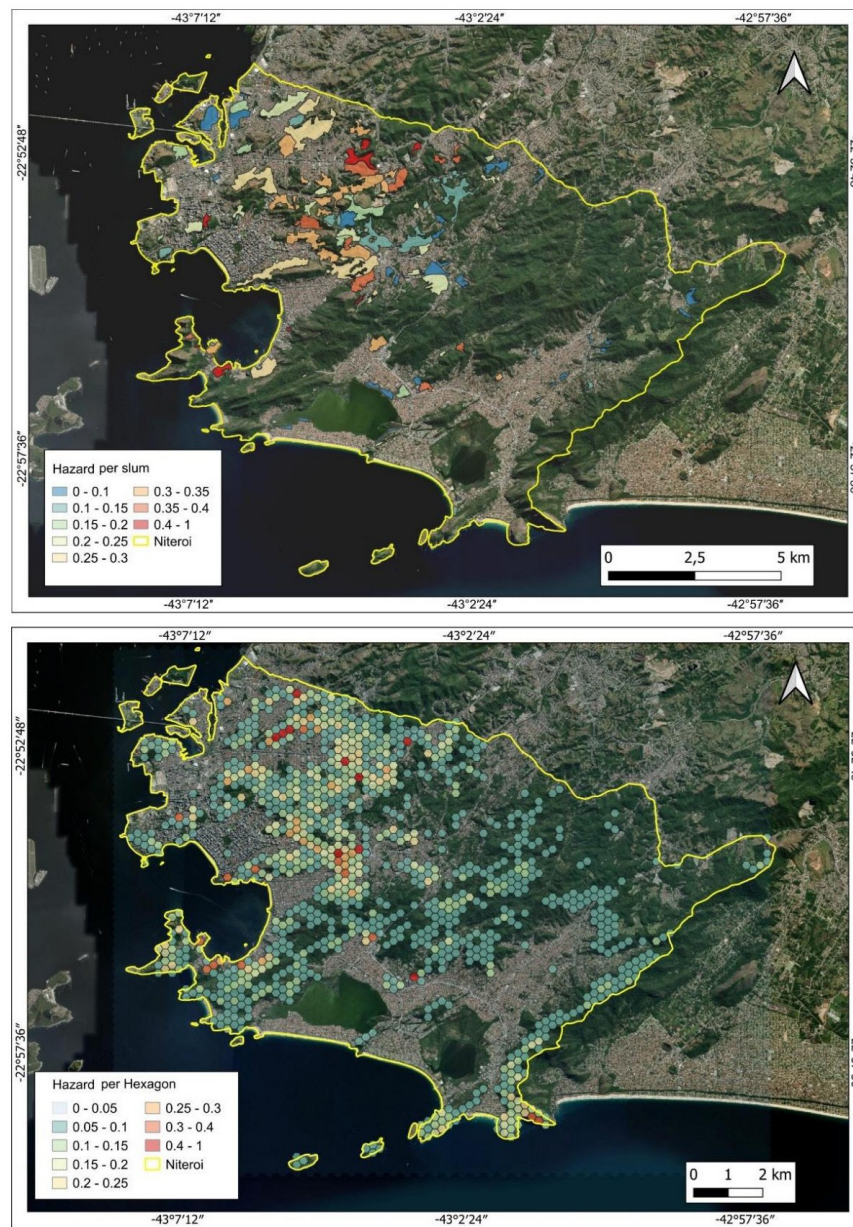


Figure 5. Hazard per slum (top) and regular hexagons (bottom), in detail. The north of the municipality presented the greatest number of hexagons with the highest hazard indices.

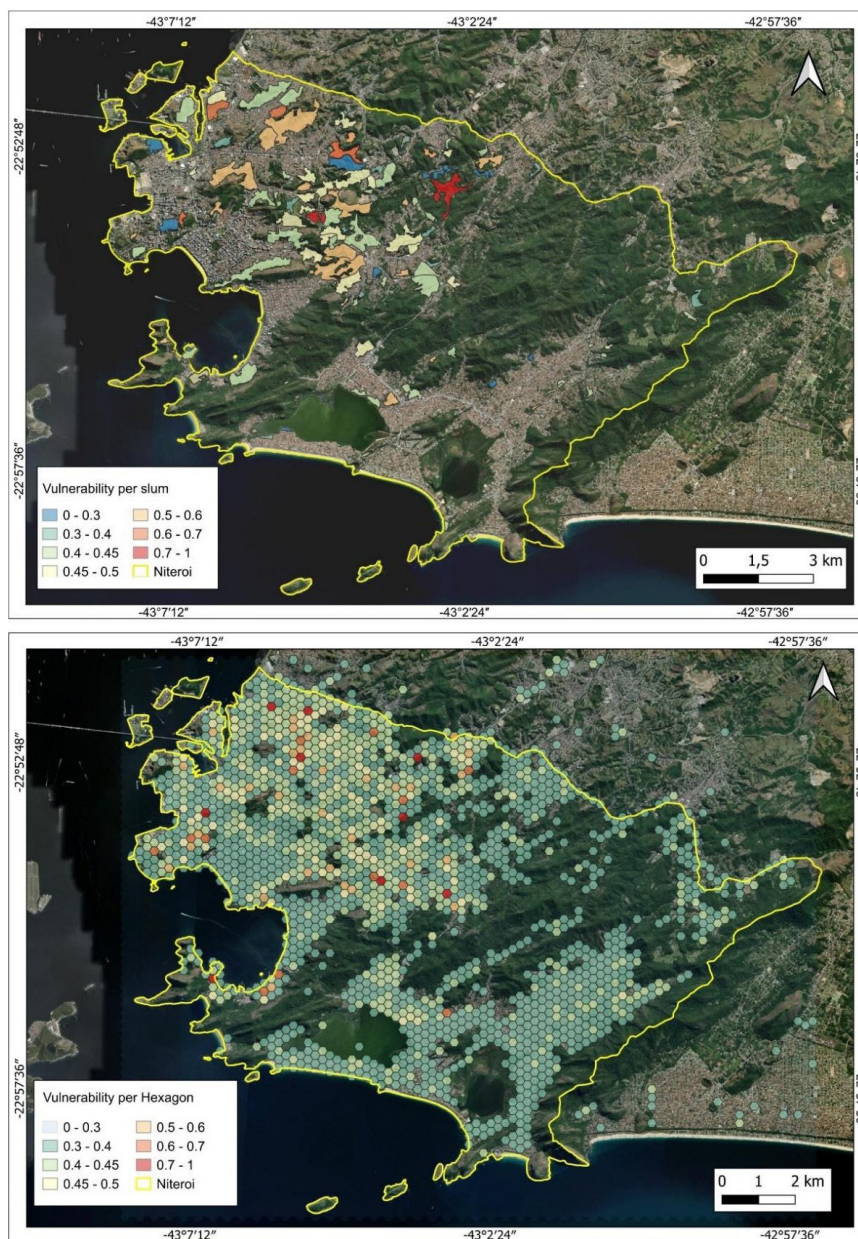


Figure 6. Vulnerability per slum (top) and regular hexagons (bottom), in detail. The vulnerability in Niterói is concentrated in the hexagons located in the northern region.

Risk index applied to Niterói

The applied method resulted in the priority ranking of 14 areas to be attended during the subsequent stages of the PMRR, with an IR threshold of 0.20 for the slums and 0.16 for the hexagons (Figure 7). The results from the calculated IR to the whole territory for both analyzed scales are presented in Figure 8. The list of priority communities defined according to the IR is shown in Table 3. After the IR ranking of these areas, participatory risk mapping was carried out with the residents of these locations, as a planned stage of the PMRR. These participatory risk mapping, as well as the risk assessment and risk mapping, were used to validate the method.

Following the IR list of prioritized high-risk areas, the other stages consisted of participatory mapping and validation of the IR. During the participatory mapping, maps of the slum and hexagons were presented to the community residents so they could indicate the highest-risk areas within their territory, according to their perception. The hexagons with the highest IR were validated with the residents, who identified that the areas of highest risk matched the hexagons with the highest IR within their slum territory. The participatory mapping was followed by detailed technical risk assessment and mapping to create and classify risk sectors. The completion of both participatory and technical risk mapping demonstrated that these assessments in the field are vital steps for the development of an adequate risk management plan.

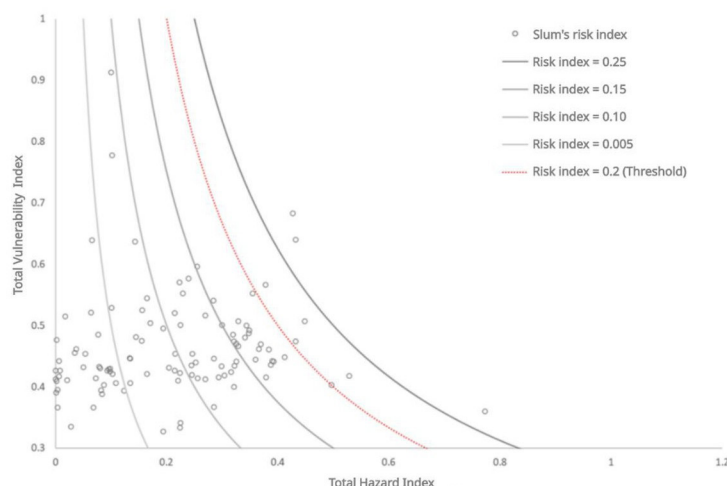


Figure 7. Total Vulnerability Index versus Total Hazard Index for Slums. The curves represent different levels of risk, with the risk threshold (0.2) indicated by the dotted red line.

Table 3. Rankings of Priority slums chosen based on Risk Index (IR).

Priority Order	Slum	Risk Index	Hexagons per Slum
1	Cova da Onça	0.35	1
2	Castro Alves	0.31	1
3	Vila Ipiranga	0.31	2
4	Bonfim	0.28	1
5	Caramujo (Dona Zinha)	0.27	1
6	Boa Esperança	0.24	1
7	Palmeira/Santo Cristo	0.24	1
8	Teixeira de Freitas	0.23	1
9	Nova Brasília	0.21	1
10	Maceió	0.20	1
11	Grota do Surucucu/Igrejinha	0.20	3
12	Boa Vista	0.16	1
13	Cavalão	0.16	1
14	Caramujo (Fazendinha)	0.16	1

It highlighted the potential of the method presented here, while also revealing limitations to the method that require attention to ensure satisfactory results.

The analysis of the IR ranking compared to the risk assessment and mapping products shows consistency between the methods, demonstrating the applicability of the presented methodology. The slums *Caixa D'Água* and *Teixeira de Freitas* serve as good examples. According to the IR index result, *Teixeira de Freitas* and *Caixa D'Água* were among the prioritized communities, ranked 2nd and 8th, respectively. Technical visits confirmed these are communities with high susceptibility and high vulnerability, with risk assessment and mapping resulting in 10 risk sectors classified as very high risk, 8 risk sectors of high risk, and 4 sectors of medium risk mapped in these slums, following the methodology outlined by Ministry of Cities (Brasil, 2024).

The mentioned slums are located in steep soil-mantled hillslopes, crossed by significant water flow lines due to the hillslope curvature. The community of *Teixeira de Freitas* has amphitheater-like slopes, crossed by multiple converging flow lines,

increasing the hazard for the slum residents spread throughout the slope (Figure 9). The *Caixa D'Água* slum encompasses a combination of flat and converging hillslopes, with medium to large flow lines crossing said slopes (Figure 10). In both slums, very high vulnerability was attested in the field, considering urban infrastructure, average income, and construction techniques.

On the other hand, slums like *Boa Esperança* demonstrated the limitations of the method, which require attention to yield satisfactory results (Figure 11). When mapped on-site, *Boa Esperança* Slum had only 3 sectors identified as medium risk, despite being ranked 6th in the list of priority slums according to the IR value.

Boa Esperança slum, in 2018, endured a landslide induced by human activity that resulted in 15 deaths and the interdiction of dozens of homes (Amaral et al., 2018). This disaster led to a spike in the Civil Defense records for this locality. The high number of occurrences recorded within the slum, although related to the same event, was captured by the index, which caused *Boa Esperança* to enter the IR ranking of priority slums. Another weakness revealed by the *Boa Esperança* community is the need

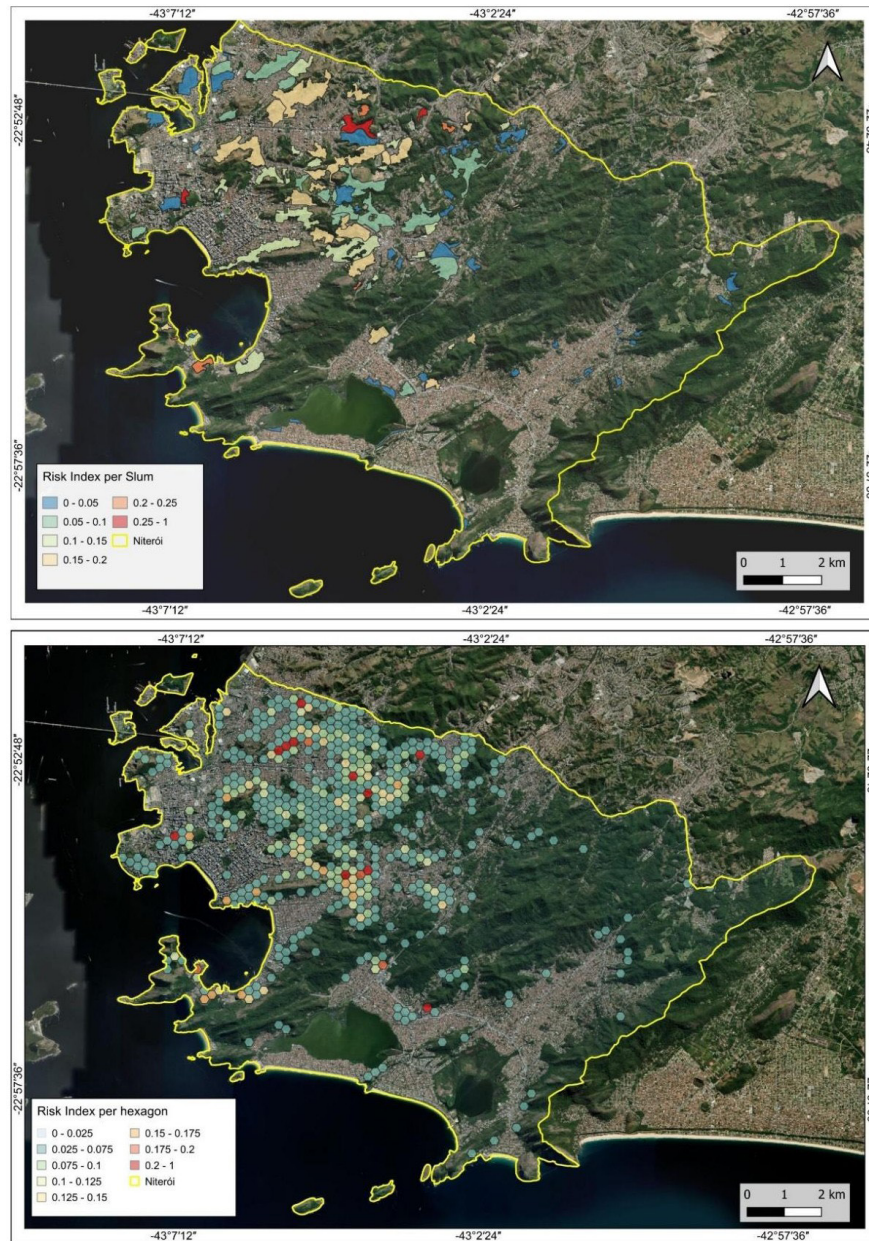


Figure 8. Risk index per slum (top) and hexagons (bottom), in detail. Reflecting the hazard index and the vulnerability index, the northern part of the municipality presents the greatest number of hexagons with the highest risk index.

to use the latest up-to-date records of structural interventions carried out by the responsible agencies. Due to this large-scale disaster in the Boa Esperança locality, several slope stabilization works were carried out to mitigate the risk in the landslide area and vicinity (Figure 11). Although working with the updated available version of the record of structural works, provided by the municipality government itself, the team only became aware of the existence of these stabilization structural works during the technical visit to the site. According to the established criterion for the IR ranking, the hexagon containing Boa Esperança stabilized territory should have been removed from the priority list, based on the understanding that the highest-risk sectors in that polygon had already been addressed.

These are significant limitations of the method, which require the user to perform a thorough filtering process of the input data to the indices, to identify and resolve such problems. It is important to mention that the I_{LR} and I_I indices (from the civil defense) should be applied with caution, as each civil defense agency has its own registry methods. Ideally, civil defense records should undergo a standardization process to facilitate scientific progress in disaster risk management. It is also crucial to ensure that the most up-to-date version of the structural interventions record is considered for the analysis.

Field observations revealed satisfactory performance of the IR for preliminary spatially distributed assessment of the territory, helping to identify high-risk areas by quantifying the

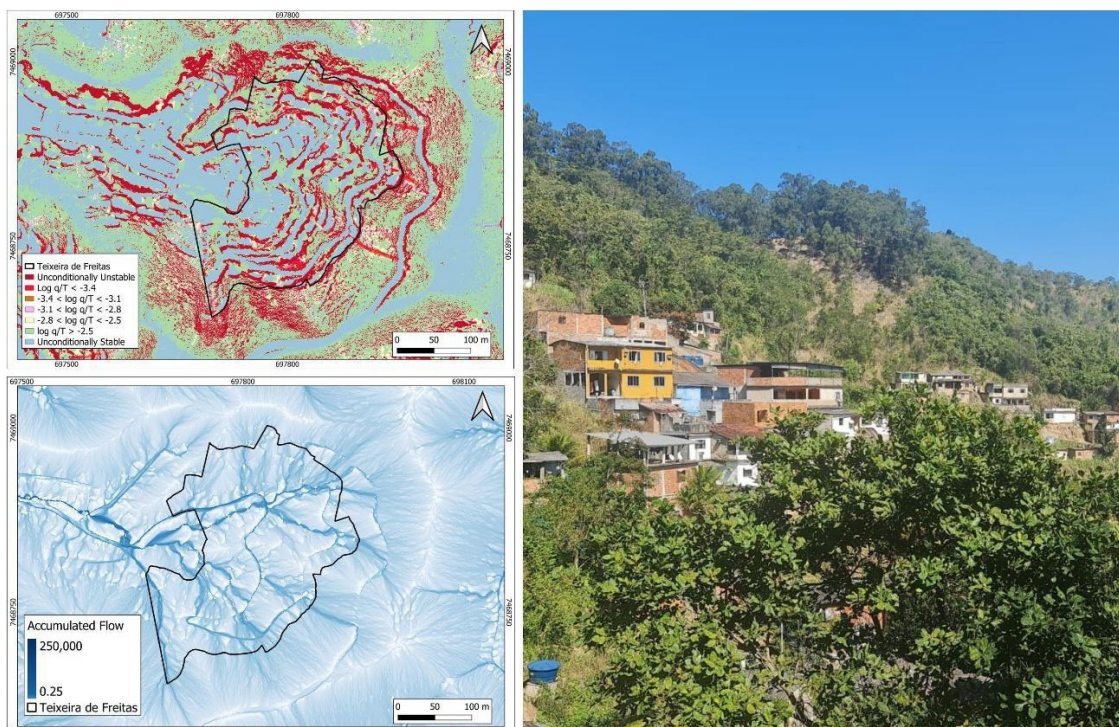


Figure 9. SHALSTAB stability map, accumulated flow map, and view of a hillslope in Teixeira de Freitas Slum. The SHALSTAB model indicates a high susceptibility to shallow landslides in warm colors along the amphitheater-like slope. The accumulated flow map shows significant convergent flow lines crossing the hillslopes.

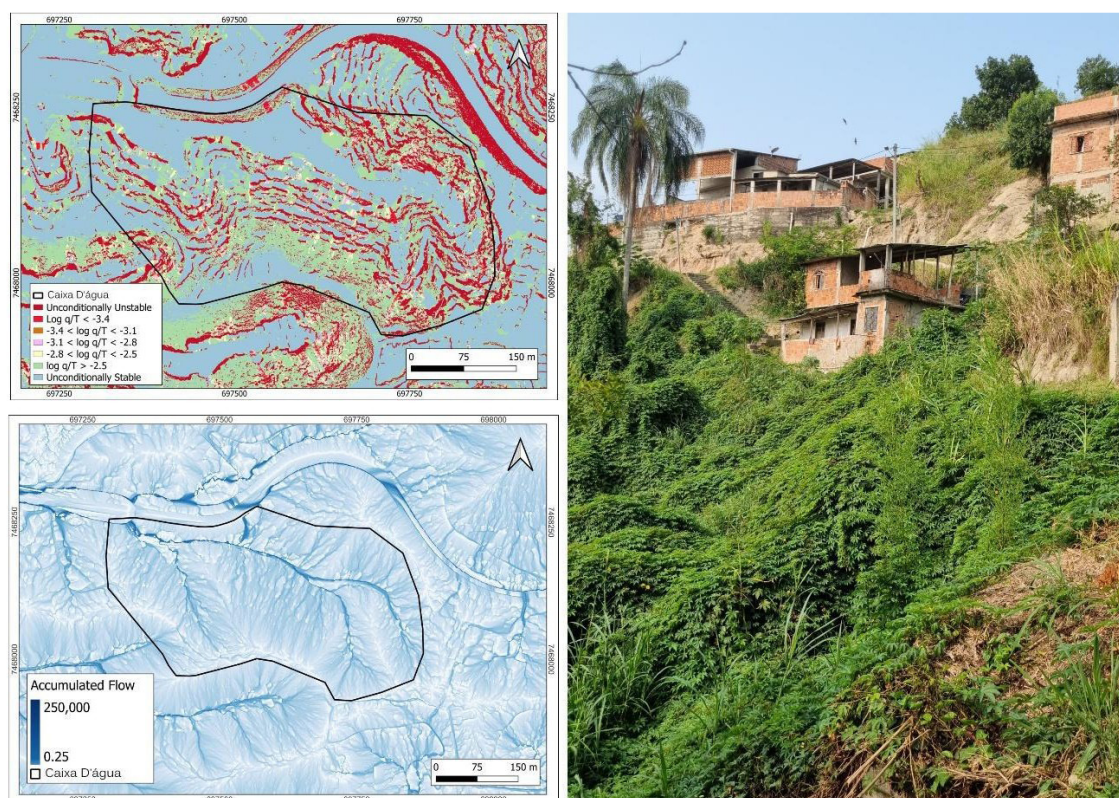


Figure 10. SHALSTAB stability map, accumulated flow map, and view of a hillslope in Caixa D'Água Slum. The SHALSTAB model indicates a high susceptibility to shallow landslides in warm colors along the hillslopes. The accumulated flow map shows significant flow lines crossing the several hillslopes of the Slum.

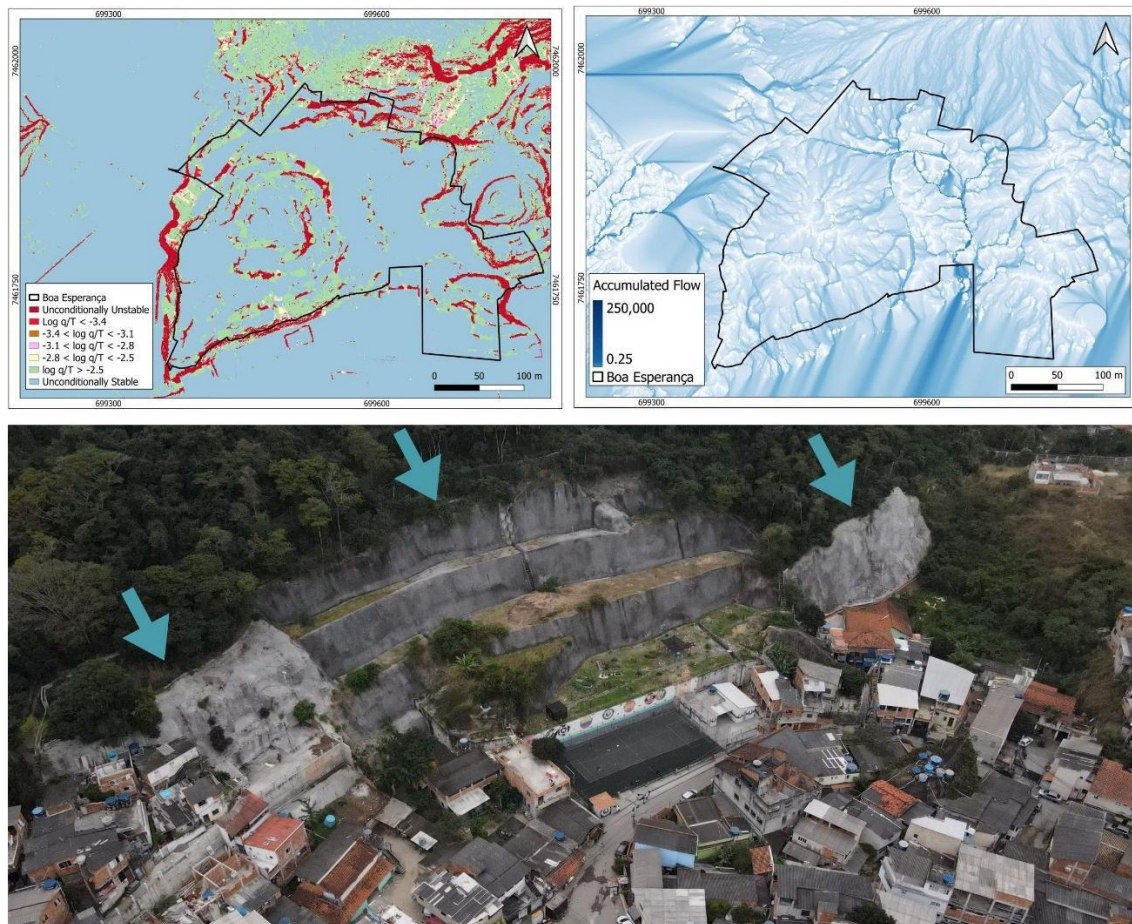


Figure 11. SHALSTAB stability map, accumulated flow map, and view of stabilization works (blue arrows) on Boa Esperança hillslope. The SHALSTAB model indicates localized areas of high susceptibility to shallow landslides in warm colors within the Boa Esperança Slum hillslope, where structural works have been carried out. The accumulated flow map shows minor to medium flow lines crossing the hillslopes of the Slum.

vulnerability and hazard susceptibility in a technical and objective manner. However, the PMRR experience further reinforces the understanding that on-site mapping and risk assessment are indispensable and must be carried out to ensure the identification of risks, with adequate detail for the coherent implementation of disaster risk planning and management.

CONCLUSION

Scientific advances in methods used to identify and prioritize high-risk areas are increasingly necessary to guide decision-makers to the application of resources based on technical arguments. Risk mapping also helps vulnerable communities to prepare themselves for managing existing risks. The application of the method presented here yielded promising results, optimizing the operational capacity of the risk assessment team. This method enables slums facing a higher risk to be mapped and have structural and non-structural interventions prioritized in the municipality's risk management planning. The methodology proved to be suitable, with the successful employment of the modeling of topographic

attributes to evaluate stability combined with the record of observed landslide data to identify hazardous areas. The hazard evaluation is further integrated with simple yet robust vulnerability data for risk assessment. The application of a multi-scale approach, particularly the application of a regular grid, is noteworthy, allowing the uniform analysis of risk within the whole territory, resulting in a detailed and spatially distributed risk analysis.

DATA AVAILABILITY STATEMENT

Civil Defense/Niterói landslide and occurrence records that support the findings of this study are available from Prefeitura Municipal de Niterói. Restrictions apply to the availability of these data, which were used under license for this study.

Vulnerability data was provided by Prefeitura Municipal de Niterói. The vulnerability data are not publicly available due to privacy restrictions.

Digital Elevation Model provided by Prefeitura Municipal de Niterói are openly available on: <https://www.sigeo.niteroi.rj.gov.br/> (accessed on 2025, January 10).

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Authors contributions

Franciele Zanandrea: Conceptualization (equal), data curation (equal), formal analysis (equal), investigation (equal), methodology

(equal), project administration (equal), supervision (equal), writing - original draft (equal), writing - review & editing (equal).

Gean Paulo Michel: Conceptualization (equal), data curation (equal), formal analysis (equal), investigation (equal), methodology (equal), software (equal), validation (equal), writing - review & editing (equal).

Aimée Guida: Conceptualization (equal), data curation (equal), formal analysis (equal), investigation (equal), methodology (equal), validation (equal), visualization (equal), writing - original draft (equal), writing - review & editing (lead).

Ivanovich Lache Salcedo: Conceptualization (equal), data curation (equal), formal analysis (equal), methodology (equal), software (equal), validation (equal), visualization (equal), writing - review & editing (equal).

Danúbia Teixeira Silva: Investigation (equal), validation (equal), visualization (equal), writing - original draft (equal), writing - review & editing (equal).

Camila Athanázio de Azevedo: Data curation (equal), investigation (equal), validation (equal), visualization (equal), writing - original draft (equal), writing - review & editing (equal).

Lucas da Silva Santos: Data curation (equal), investigation (equal), software (equal), validation (equal), visualization (equal), writing - review & editing (equal).

Isadora Tschiedel Thys: Conceptualization (equal), data curation (equal), formal analysis (equal), investigation (equal), writing - review & editing (equal).

Vanessa Caldas Cardoso: Conceptualization (Equal), data curation (equal), investigation (equal), validation (equal), writing review & editing (equal).

Editor-in-Chief: Adilson Pinheiro

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